

# Project Design Phase: Solution Architecture

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|----------------|------------------------------------------------|
| Date:          | 04 October 2026                                |
| Team ID:       | SWTID-2026-5601                                |
| Project Name:  | Auto Ticket Classification using Flow Designer |
| Maximum Marks: | 4 Marks                                        |

## Solution Architecture Overview

Solution architecture bridges the gap between business problems and technology solutions [cite: 22]. For the 'Auto Ticket Classification using Flow Designer' project, the architecture defines the structure, behavior, and data flow of the automated ServiceNow IT service management system.

## Architecture Components & Data Flow

| Component Layer             | Architecture Element                    | Description & Data Flow                                                                                                                                   |
|-----------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presentation Layer          | ServiceNow Service Portal & UI          | Provides the user interface for students and faculty to submit support tickets with short descriptions, detailed descriptions, and attachments.           |
| Integration & Trigger Layer | Database Insert Trigger / Business Rule | Detects newly created incident records in the Task database and asynchronously fires the Flow Designer execution engine.                                  |
| Processing & Triage Layer   | Flow Designer & Decision Tables         | Executes automated text parsing, keyword extraction, and decision matrix matching against incident fields to predict category, subcategory, and priority. |
| Routing & Storage Layer     | Assignment Engine & CMDB Database       | Automatically assigns the ticket to the correct technical assignment group (e.g., Network, Hardware, Software)                                            |

|                              |                                         |                                                                                                                                                                                                |
|------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notification & Logging Layer | Email Engine & CloudWatch / System Logs | and persists data in ServiceNow CMDB.<br><br>Dispatches automated email notifications to end-users and support teams while logging transaction states for auditing and performance monitoring. |
|------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Solution Architecture Data Flow Diagram (Structured Representation)

The following flow outlines the end-to-end architecture pipeline corresponding to the project data flow:

- **1. User Submission:** End-user creates an incident via Service Portal -> Submits Short & Detailed Description.
- **2. Event Trigger:** New record insert triggers ServiceNow Flow Designer workflow pipeline.
- **3. Automated Triage:** Flow Designer parses text -> Evaluates Decision Tables & Keyword Matchers.
- **4. Assignment & Routing:** System assigns Category, Priority -> Automatically routes to appropriate Assignment Group.
- **5. Notifications & Output:** Dispatches confirmation emails to user -> Updates Agent Dashboard & Audit Logs.

SOLUTION ARCHITECTURE: AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

